

Sheet 3

Moment methods for differential and integral equations

- 1- Use the method of moments to get approximate solution for the boundary value problem,

$$-\frac{d^2 \phi}{dt^2} = 1 + 4t^2, \quad \phi(0)=0, \quad \phi(1)=0$$

Represent ϕ using Triangle functions T_n , $n=1,3$ and use W_n = piecewise constant functions P_n .

- 2- Use the method of moments to get approximate solution for the potential V across the line $0 \leq x \leq 1$ satisfied by Poisson' equation,

$$\frac{d^2 V}{dx^2} = 1 + x - 2x^2, \quad V(0) = V(1) = 0$$

Represent V using Triangle functions T_n , use weighting function as using piecewise constant P_n and let $N=4$. (Get solution matrix).

- 3- Use the method of moments to get approximate solution for the boundary value problem,

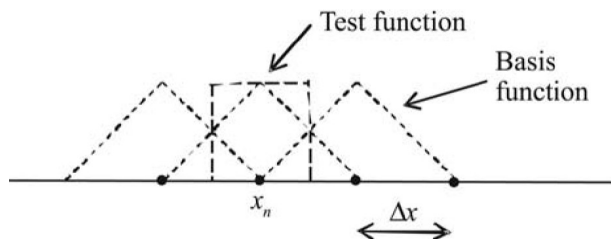
$$-\frac{d^2 \phi}{dx^2} + 2 \frac{d\phi}{dx} = 1 + 3x, \quad \phi(0)=0, \quad \phi(1)=0$$

Represent ϕ using Triangle functions T_n , $n=1,3$ and use W_n = piecewise constant functions P_n . and let $N=4$. (Get solution matrix).

- 4- Consider the following one-dimensional scalar Helmholtz equation:

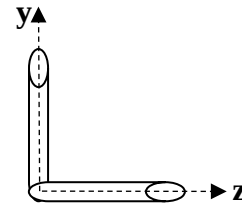
$$\frac{d^2 E_y}{dx^2} + k^2 \frac{dE_y}{dt} = g(x), \quad 0 \leq x \leq 1$$

using triangular basis functions for E_y . Enforce the differential equation with pulse testing functions such that basis functions straddle two cells of dimension Δx , while the test functions are defined between the centers of adjacent cells as shown in the Figure. Determine the mn^{th} entry of the resulting matrix.



5- (Get the solution matrix)

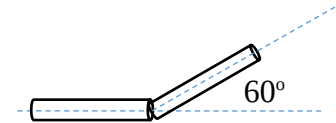
A metallic wire of length 1 m and diameter $D = 2 \text{ mm}$ is bended to form L- shape as shown in the figure. If its potential is 4 volt. Use method of moments to determine the linear charge distribution on the wire by dividing the length into 4 equal segments. Take the basis functions as pulses (piecewise constant).



6- A metallic wire of length $L = 8 \text{ cm}$ and diameter $D = 2 \text{ mm}$ is

bended as shown in the figure. If it's potential is 4 volt. Use method of moments to determine the linear charge distribution on the wire.

(Divide the length into 4 equal segments and take the basis functions as pulses (piecewise constant)).



7- (Get the solution matrix)

A square conducting plate of side $2a$ and neglected thickness is placed on the $z=0$ plane with center at origin. The capacitance of the plate is defined as,

$$C = \frac{Q}{V} = \frac{\int_{-a}^a \int_{-a}^a \sigma(x,y) dx dy}{V}$$

where $\sigma(x,y)$ is the surface charge density and V is its potential.

Use the moments method to get $\sigma(x,y)$ at the centers of 2×2 divisions ($N=4$) of the plate and Calculate $C/2a$ in this case (in terms of the solution matrix).

[use subsectional functions $f_n = 1$ on Δs_n and $f_n = 0$ on all other S_o , and use weighing functions

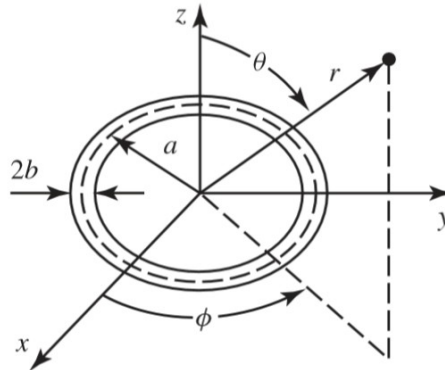
$$W_n = \delta(x-x_n) \delta(y-y_n)]$$

8 - A long wire carrying AC current of amplitude $I = 1.0 \text{ A}$ and frequency 3 GHz , is located at height $h = 0.5 \text{ cm}$ above a long metallic strip of width $w = 20 \text{ cm}$ as shown in the figure.

Use integral equation method to get the induced current on the metallic strip. Express this current using piecewise constant functions P_n , (Take $N = 4$, assume thickness very small $\ll \lambda$, get solution matrix)



9- A circular loop of radius $a = 0.2\text{m}$ is constructed out of a wire of radius $b = 10^{-3}\text{m}$, as shown in Figure P12-1. The entire loop is maintained at a constant potential of 1V . Using integral equation techniques, determine and plot for $0^\circ \leq \phi \leq 360^\circ$ the surface charge density on the wire. Assume that at any given angle the charge is uniformly distributed along the circumference of the wire.



10- Repeat Problem 8 when the loop is split into two parts; one part (from 0 to 180°) is maintained at a constant potential of 1V and the other part (from 180 to 360°) is maintained at a constant potential of 2V .

11- 4-a A metallic ring of radius $a = 1\text{m}$ is kept at potential 1 volt , Use method of moments to determine

the charge charge distribution by dividing the ring into 4 equal segments. Compute the charge

distribution at the center of the segments.

Take the basis functions as pulses (piecewise constant).

